Lindsey C. Bowen

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Education

The University of Alabama — Tuscaloosa, AL

M.S. in Electrical Engineering, Accelerated Master's Program — *Expected Dec. 2027*

B.S. in Computer Engineering; Minors in Mathematics and Computer Science — *Dec. 2026*

- Undergraduate GPA: 3.8/4.0, Magna Cum Laude
- Society of Women Engineers, IEEE, Mentor UPP, UA CAD Club

Experience

Correll Technologies — Remote

Full-Stack Software Engineering Intern — *May 2026 – Present*

- Build UI features and API webhooks for a healthcare platform with TypeScript, React, Next.js, Node.js, and PostgreSQL
- Integrate cloud-hosted services and authentication with AWS and Microsoft Entra ID
- Develop, debug, and ship application features through a Jira-based team workflow

University of Alabama, CLAWS Laboratory — Tuscaloosa, AL

Undergraduate Research Assistant — *Aug. 2025 – Present*

- Develop an STM32-based dual-LIS2MDL magnetometer sensor to detect vape coil activation and characterize puffs
- Sample at 20 Hz via dual timer interrupts, averaging two offset-cancelled readings per 50 ms window to reduce noise
- Cut baseline noise from ∼255 to ∼9 mG (96%) with a 9-parameter ellipsoid calibration; wrote the team's test procedure
- Implement RF edge detection, USB CDC, RTC timestamps, FATFS SD logging, and double buffering in embedded C
- Run aerosol experiments (mass loss, particle count, temperature, power, puff interval) with a Raspberry Pi rig and CPC
- Analyze data in Python and contribute vape topology analysis to a conference paper on e-cigarette aerosol dynamics

EOSYS Group — Decatur, AL

Systems Integration Co-op Engineer — *May 2024 – Aug. 2024; Jan. 2025 – Aug. 2025*

- Configured several hundred Foxboro I/A and PLC I/O tags to build working DCS subsystems for industrial clients
- Developed XML-based automation tools for managing large industrial control-system configuration datasets
- Designed and tested HMI interfaces for Schneider Electric Foxboro control systems
- Designed a factory-acceptance testbed with redundant Foxboro CPs, FBM233 modules, network switches, and fiber converters

University of Alabama — Tuscaloosa, AL

Cleanroom Research Assistant — *Aug. 2025 – May 2026*

- Performed and trained others in photolithography and spin coating; presented wafer work at UA's research conference

Selected Projects

Electronic Nose for Breath-Biomarker Detection — Embedded Systems, PCB Design, BLE

- Designed a portable sensor system for acetone (0–10 ppm) and breath-range CO_2, H_2, and ethanol
- Led acetone and ethanol sensing, including mole-fraction and ppm concentration calculations for sensor testing
- Built KiCad PCBs with Wheatstone-bridge conditioning and differential ADCs; streamed data over BLE from an nRF52840
- Modeled the full device in Fusion 360, including the enclosure and breath-flow chamber

Precision Thermistor Sensor for Hot Beverages — Analog Design, PCB Design | Fall 2026

- Designing a thermistor-bridge and instrumentation-amplifier front end targeting ±0.05 °C accuracy
- Laying out a custom KiCad PCB pairing a Seeed XIAO nRF52840 with a 16-bit MCP33131 SAR ADC over SPI

Leadership

Electrical and Computer Engineering Department Ambassador — University of Alabama

- Represent the department at prospective-student events, tours, and outreach activities

IEEE Secretary — University of Alabama Chapter

- Plan and publicize IEEE events and lead joint meetings with HKN and the UA CAD Club

Technical Skills

Programming: C, C++, Python, TypeScript, JavaScript, SQL, Java, SystemVerilog
Embedded: STM32, nRF52840, ESP32, SPI, I2C, UART, USB CDC, timers, interrupts, FATFS, BLE
Software: React, Next.js, Node.js, Git, PostgreSQL, Tailwind, AWS
Engineering Tools: KiCad, STM32CubeIDE, LTspice, Fusion 360, Vivado, oscilloscopes, THT/SMD soldering, PCB assembly
Laboratory: Aerosol instrumentation, experimental design, photolithography, spin coating, microfabrication